Symmetry

Year 4 · **Lesson Worksheet**

Worked Examples

Lines of Symmetry: How many lines of symmetry does a regular pentagon have?

Answer: A regular pentagon has 5 sides, so it has 5 lines of symmetry (one through each vertex and the midpoint of the opposite side).

Reflecting a Point: Reflect the point (3, 5) in the y-axis (the line x = 0). Give the coordinates of the image.

Answer: Reflecting in the y-axis keeps the y-coordinate the same and changes the sign of the x-coordinate: the image is (−3, 5).

Order of Rotational Symmetry: What is the order of rotational symmetry of a rectangle that is not a square?

Answer: A rectangle looks the same twice during a full 360° turn (at 0° and 180°), so its order of rotational symmetry is 2.

Symmetry Basics (15 questions)

1. How many lines of symmetry does a square have?
2. How many lines of symmetry does an equilateral triangle have? A) 1 B) 2 C) 3 D) 4 E) 5
3. How many lines of symmetry does a regular hexagon have?
4. How many lines of symmetry does a rectangle (that is not a square) have? A) 0 B) 1 C) 2 D) 3 E) 4
5. How many lines of symmetry does the capital letter T have?
6. How many lines of symmetry does the capital letter H have?
7. Which of these capital letters has exactly 1 line of symmetry? A) H B) O C) N D) A E) S
8. How many lines of symmetry does an isosceles triangle have (a triangle with exactly two equal sides)?
9. How many lines of symmetry does a regular octagon have?
10. Select ALL that apply: which of these shapes have at least one line of symmetry? A) Square B) Scalene triangle C) Rectangle D) Parallelogram (not a rectangle or rhombus) E) Regular pentagon
11. A shape has a single vertical line of symmetry down its middle. If the left half has an area of 18 cm^2, what is the total area of the shape?
12. Reflect the point (2, 7) in the x-axis (the line y = 0). Give the coordinates of the image.
13. Reflect the point (4, 3) in the y-axis (the line x = 0). What are the coordinates of the image? A) (4, −3) B) (−4, 3) C) (−4, −3) D) (3, 4) E) (4, 3)
14. How many lines of symmetry does a rhombus have (assuming it is not a square)?
15. A kite shape has how many lines of symmetry?

Symmetry & Reflection (15 questions)

1. What is the order of rotational symmetry of a square?

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2. What is the order of rotational symmetry of an equilateral triangle?

3. What is the order of rotational symmetry of a regular hexagon? A) 2 B) 3 C) 4 D) 5 E) 6

4. What is the order of rotational symmetry of a parallelogram that is not a rectangle?

5. A regular pentagon has rotational symmetry. What is its order of rotational symmetry, and how many lines of symmetry does it have?

6. Which of these shapes has rotational symmetry of order 2 but NO lines of symmetry? A) Rectangle B) Rhombus C) Parallelogram (not a rectangle or rhombus) D) Square E) Isosceles trapezium

7. Reflect the point (5, 2) in the line y = x. Give the coordinates of the image.

8. Reflect the point (−3, 4) in the x-axis. Give the coordinates of the image.

9. Select ALL that apply: which of these capital letters have rotational symmetry of order 2? A) H B) N C) F D) S E) Z

10. A shape drawn on a grid has vertices at (1, 1), (1, 5) and (3, 5) on one side of a vertical mirror line x = 6. To complete the full symmetric shape, find the coordinates of the reflected image of the vertex (1, 1).

11. A rectangle measuring 8 cm by 3 cm has how many lines of symmetry in total? A) 0 B) 1 C) 2 D) 3 E) 4

12. A regular shape has 9 lines of symmetry. How many sides does it have, and what is its order of rotational symmetry?

13. The point (6, −2) is reflected in the horizontal line y = 1. Find the coordinates of the image.

14. A shape has exactly one line of symmetry and no rotational symmetry (other than the trivial order 1). Name one example of such a shape from: kite, square, rectangle, regular pentagon.

15. A shape has both line symmetry and rotational symmetry of order greater than 1. Which of these is NOT an example? A) Square B) Regular hexagon C) Rectangle D) Scalene triangle E) Rhombus

Advanced Symmetry Challenge (15 questions)

1. A regular polygon has rotational symmetry of order 8. How many sides does it have, and how many lines of symmetry does it have?

2. A shape has 5 lines of symmetry and rotational symmetry of order 5. Which shape could it be? A) Square B) Regular pentagon C) Rectangle D) Rhombus E) Parallelogram

3. The point (4, −5) is reflected in the x-axis, and then the image is reflected again in the x-axis. What are the final coordinates?

4. A pattern is made by reflecting a right-angled triangle in a vertical mirror line. The original triangle has vertices at (0,0), (3,0) and (0,2), and the mirror line is x = 3. Find the coordinates of the reflected image of the vertex (0, 0).

5. Select ALL that apply: a point P is reflected first in the x-axis, then the image is reflected in the y-axis. If P = (4, 3), which of the following are true about the final image? A) The final image is (−4, −3). B) The final image could also be reached by a single 180° rotation of P about the origin. C) The final image is (4, −3). D) The x-coordinate of the final image is negative. E) The y-coordinate of the final image is positive.

6. A shape has point symmetry (rotational symmetry of order 2) but no line symmetry. Its area is 36 cm². After a 180° rotation about its centre, what is the area of the image?

7. How many lines of symmetry does a regular 9-sided polygon (nonagon) have? A) 6 B) 7 C) 8 D) 9 E) 10

8. The point (−3, 2) is reflected in the line x = 2. Find the coordinates of the image.

9. A square of side 8 cm has its two diagonals drawn, together with a vertical and a horizontal line through its centre. How many lines of symmetry pass through the centre of the square in total?

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10. A shape has rotational symmetry of order 1. What does this mean? A) It has no rotational symmetry (it must be turned through a full 360° before it looks the same again) B) It has exactly 1 line of symmetry C) It can be rotated once to look the same D) It has infinite rotational symmetry E) It is impossible for a shape to have order 1

11. A regular hexagon has centre (0, 0) and one vertex at (5, 0). Using the hexagon's rotational symmetry of order 6, through what angle, in degrees, must it be rotated (about its centre) to map the vertex (5, 0) onto the next vertex around?

12. A shape with 4 lines of symmetry also has rotational symmetry. A student claims its order of rotational symmetry must be exactly 2. Is the student correct? Give the correct order of rotational symmetry for a regular shape with 4 lines of symmetry.

13. Which of these shapes has NO lines of symmetry AND no rotational symmetry (order 1 only)? A) Parallelogram (not a rectangle or rhombus) B) Scalene triangle C) Rectangle D) Letter S E) Isosceles triangle

14. A point at (6, 2) is reflected in the line y = x to give point Q. Point Q is then reflected in the x-axis to give point R. Find the coordinates of R.

15. Select ALL that apply: which of these shapes have at least one line of symmetry AND rotational symmetry greater than order 1? A) Square B) Regular hexagon C) Rectangle D) Scalene triangle E) Parallelogram (not a rectangle or rhombus)

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